VEHICLE HEALTH ANALYSIS AND SPARE PARTS E-COMMERCE

A PROJECT REPORT

Submitted by

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ABSTRACT

Vehicles need constant servicing and repairs after a certain interval of time and if not repaired at fixed intervals it can lead to accidents. The objective of the project is to give an idea to the owner about the estimated insurance claim which he/she can get in case of certain faults and issues whether the insurance claim exists for the vehicle or not. There are many automobile health monitoring systems which are used to detect and diagnose faults and issues like On-Board Diagnostic (OBD) and Engine Control Units (ECU) which tells the owner where the fault has occurred. In this project, we have used Support Vector Regression, TFIDF, LSTM and a dataset containing the car prices to train a machine learning algorithm that can predict the estimated insurance claim based on the faults which the owner has entered.

The main purpose of this system is to reduce the amount of scams that takes place when the owner has no idea about the actual cost. This project aims to mitigate it by helping the owner to know about their vehicle's full insurance which they can rightfully claim and the amount they are required to pay.

Keywords — On-Board Diagnostic, Engine Control Units, Support Vector Regression, TFIDF, LSTM

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Avinash reddy Vasipalli

Yuvraj Singh Chauhan

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CHAPTER 1

Introduction

Many of today's vehicles include a wide range of systems and components that perform various operations while the vehicle is in use. Over time, repeated use of the vehicle may cause failure of individual systems or components. The failure of one component or system affects the overall performance of vehicle. For continuous better performance of vehicle, the best way is to carry out prescribed periodical inspection and repair in a prescribed way. It requires years of experience to inspect manually, thus the electronic vehicle health monitoring system is used. In order to get the maximum insurance sum of the vehicle that falls under the company's insurance claim policy, this is where vehicle insurance application comes which helps the owner to get maximum possible claim.

Insurance systems are widespread in today's world. They are being extensively deployed to check and verify the insurance sum of a particular entity. To name a few, insurance systems are used in healthcare, business, home, life and travel. Insurance systems uses the information of the various cost prices of the particular entities which they use along with the current issue in order to estimate the insurance claim. Vehicle Insurance Application uses machine learning algorithms like Support vector regression, TFIDF and LSTM to predict an estimated price of the insurance sum based on the faults given by the owner. Thus, it helps in reducing

scams like not paying the money, giving way less money than the actual amount, etc., and helps the owners to get their maximum possible claim.

A contract between an insurer and an insured under which the insurer agrees to make up any losses resulting from the risk insured against is known as an insurance policy. In exchange, the insured agrees to pay an upfront amount known as the premium. The Policy is the name of the legal document comprising the insurance contract. The objects, such as property or lives, that serve as the foundation for insurance are referred to as the subject matter of insurance. The Legislature enacted the Insurance Act of 1938 on February 26, 1968, and it went into effect on July 1, 1939. It has undergone several revisions. It encompasses all of India. It is an act to alter and combine the laws governing the insurance industry.

In practice, insurance entails everyone in the pool agreeing to shoulder a percentage of the loss. Because it is unclear who will lose money in such a deal, everyone has found it to be favorable. Thus, throughout time, the notion that such a shared resource pool should be managed by professionals who would determine the amount of the contribution to be imposed on each individual emerged. The concept of insurance emerged in this manner. Although insurance has evolved into a highly commercial endeavor in contemporary times, the basic idea has remained the same: the insurer collects premiums from a large number of people and insures them against a wide range of hazards.

Challenges

- Dataset for this project was a big problem as many sites did not have a proper arranged data sets for analysis. Some had name and value while some had distance travelled but no other attribute. In this process there was a point where insufficient data was available for the machine to work, wherein prediction was not as expected.
- After a thorough search and time, we could find a dataset that could work with the analysis where it had horse power, milage, stokes and other import elements for a vehicle's engine
- When it comes to results, with the use of liner regression it was always possible for negative values which implies they are not worth insurance or the vehicle health was bad.
- These cases occurred frequently during the data analysis and prediction time which took a lot of time
- Even though the value is predicted, with more data on vehicle and previous insurance issued for those along with extra attributes can really help the machine provide more accurate value
- Also, use of other algorithms like SVM, logistic regression, Decision tree, KNN, K-mean can provide more insights on this prediction of value.

CHAPTER 2

Literature Survey

I. GAFFAR GULAB MOMIN

"A REVIEW ON VEHICLE HEALTH ANALYSIS MONITORING SYSTEM" [2022]

Vehicles need repair after a certain interval of time and if are not repaired at fixed intervals it can lead to accidents. The objective of the project is to continuously notify the owner about the air pressure of tire, quality of exhaust gas, wear of breaks, leakage of oil, the excessive heating of the engine or the blockage in fuel pipe, etc. Tire pressure monitoring sensor is used to get live tire pressure reading, gas sensor gives the readings of amount of carbon dioxide (CO2), Nitrogen Oxides (NOx) etc. as amount and quality of exhaust gives the proper working condition of engine, flow sensor is used that senses the over flow of oil or leakage of oil, temperature sensor is employed in the engine that warns the owner about overheating of engine and a flow sensor is used that senses the proper flow of fuel in the pipe.

II. E. CHANDRASEKHAR

"VEHICLE HEALTH MONITORING SYSTEM" [2021]

The phrase "on-board diagnostics" in the automobile industry refers to a vehicle's capacity for self-diagnosis and reporting. The state of the different car subsystems may be accessed by the owner or a repair professional

using OBD systems. Since its inception in the first iterations of on-board car computers in the early 1980s, the quantity of diagnostic data accessible via OBD has changed greatly. When a fault was discovered, older versions of OBD would only turn on a "idiot light" or malfunction indication light without giving any other details. Modern OBD implementations give real-time data via a standardised digital communications interface in addition to a defined set of diagnostic problem codes that enable quick identification and correction of vehicle issues. OBD-I or OBD-II buses are often included in cars to specify reports or diagnostics of the vehicle.

III. PARMESH K, RASHMI PRAFULLAKUMAR NERIYA, GANESAN K

"VEHICLE HEALTH MONITORING AND ANALYSIS" [2015]

Modern vehicles are integrated with a greater number of ECUs and sensors which make them smarter in terms of engine decisions, performance, fuel efficiency, security and stability. ECUs also provide useful data in the form of diagnostic codes via OBDII (on-board diagnostics) protocol, these codes determine a specific issue in the vehicular system which is used by service technicians to address the issues occurred. This paper deals with the exploring the potential of OBDII codes from engine ECU and compute the data on an Android device to monitor the vehicle health status in real-time. The whole idea is to provide portable diagnosis solutions and integration of such system into in-vehicle entertainment like Google's Android Auto and Apple's CarPlay which can be one of key component for V2X and IOVs (Internet of vehicles).

IV. MING-SONG LIN, CUN-XIANG LIU

"RESEARCH OF THE SYSTEM OF REMOTING VEHICLE INSURANCE ESTIMATION" [2016]

The present approach for estimating auto insurance has a lot of issues and takes too long. With the help of the remote vehicle insurance estimation system, the insurance provider will receive a picture of the vehicle loss situation in order to assess the damage promptly, make a quick decision, expedite processing of claims, reduce operating costs, and enhance the effectiveness of managing compensation. In contrast, it is clear that the remote vehicle insurance estimation system, which is a long-range system as opposed to a vehicle insurance estimate model, is more suited to meet the demands of the market today. A significant promotion is required in the Chinese insurance industry.

V. UMA SHANKAR YADAV, RAVINDRA TRIPATHI, MANO ASHISH TRIPATHI

"REVIEW OF THE MOTOR VEHICLE INSURANCE INDUTRY IN INDIA" [2022]

The most popular and efficient way to safeguard personal property is through motor vehicle insurance, which has garnered a lot of interest. Understanding the present state of the Malaysian auto insurance market is essential. This setting serves as the foundation for this article's analysis of the Indian motor insurance sector's growth model and its connection to the country's economic growth. This article examines methods for

achieving targeted product advertising and regionally-specific motor vehicle insurance (VMI). The goal of this paper is to build a solid foundation for fostering the growth of the Indian car insurance market and its connection to the country's vehicle manufacturing. The article examines the impact of Malaysia's VMI policy on the nation's economy and motor vehicle manufacturing in comparison to other developed nations. The article also emphasizes the impact of COVID-19 on the VMI sector.

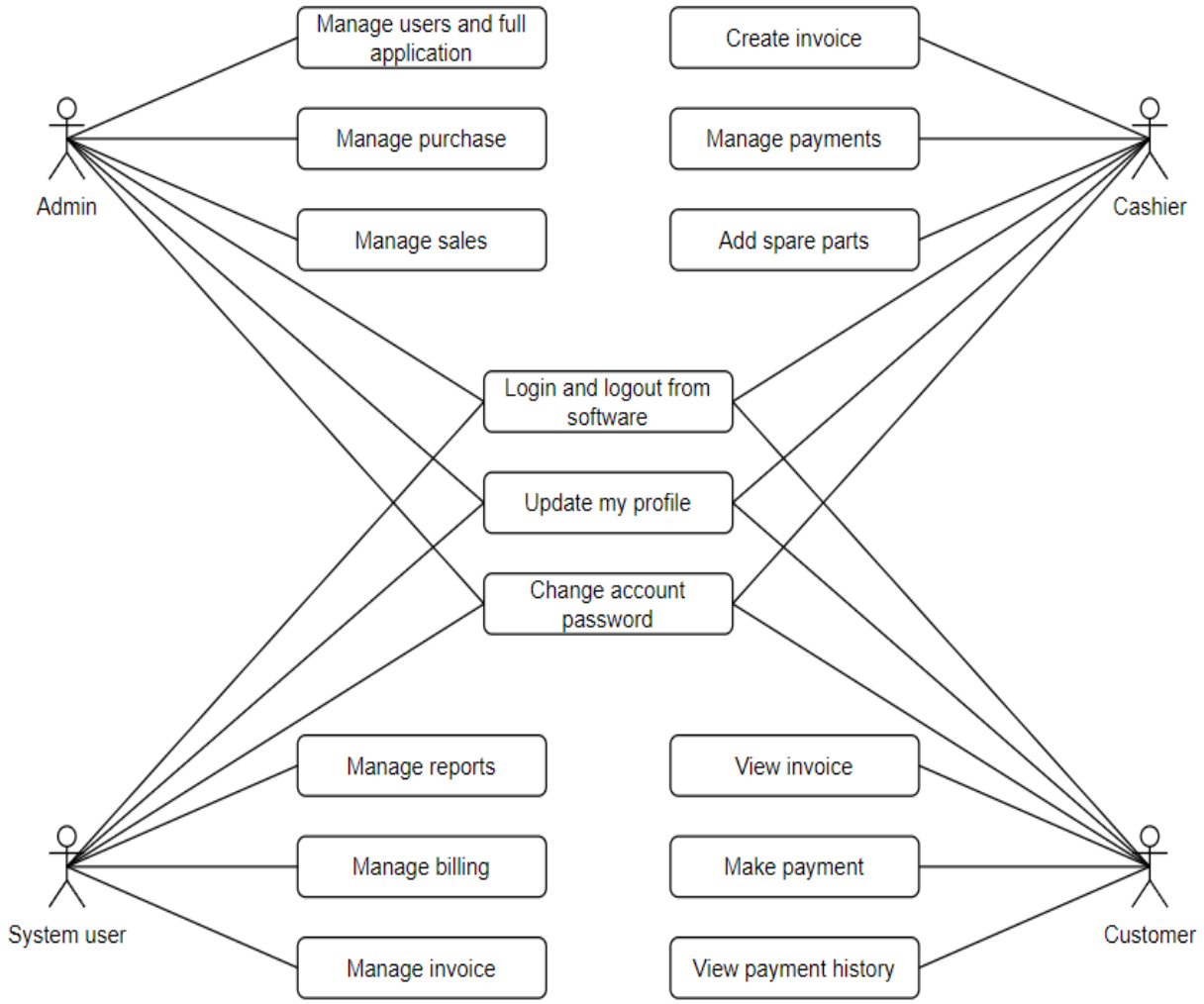
VI. JACOB MEHLMAN, JIAHUI WU, HANSHEN JING JINGYI QI

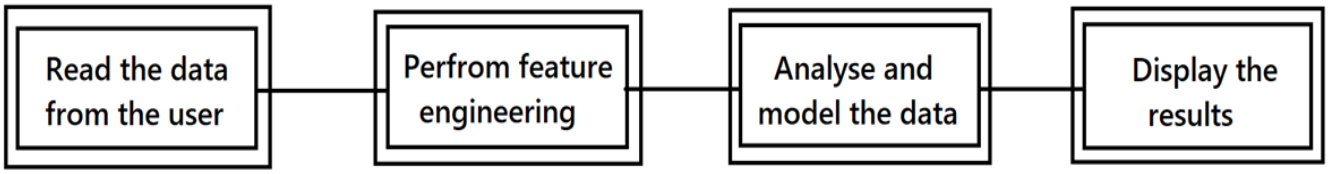
"PREDICTION OF CUSTOMER'S INTERESTS IN VEHICLE INSURANCE" [2022]

The insurance company that has provided health insurance to its customers now needs to find whether policyholders(customers) from the past year will also be interested in vehicle insurance provided by the company. In this study, we build statistical models using bagging, AdaBoost, random forest, XGBoost, and Convolutional Neural Network based on demographic information, vehicle situation, and insurance policy information to predict customers' willingness of buying vehicle insurance. We generate ROC curves, confusion matrices, and accuracy scores to compare the models and finally choose the CNN model and make recommendations to the insurance company.

CHAPTER 3

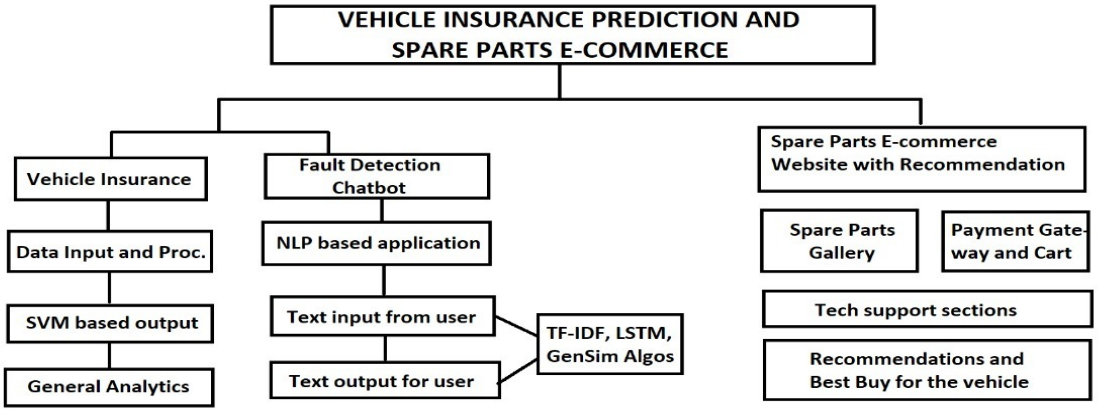
Block Diagram

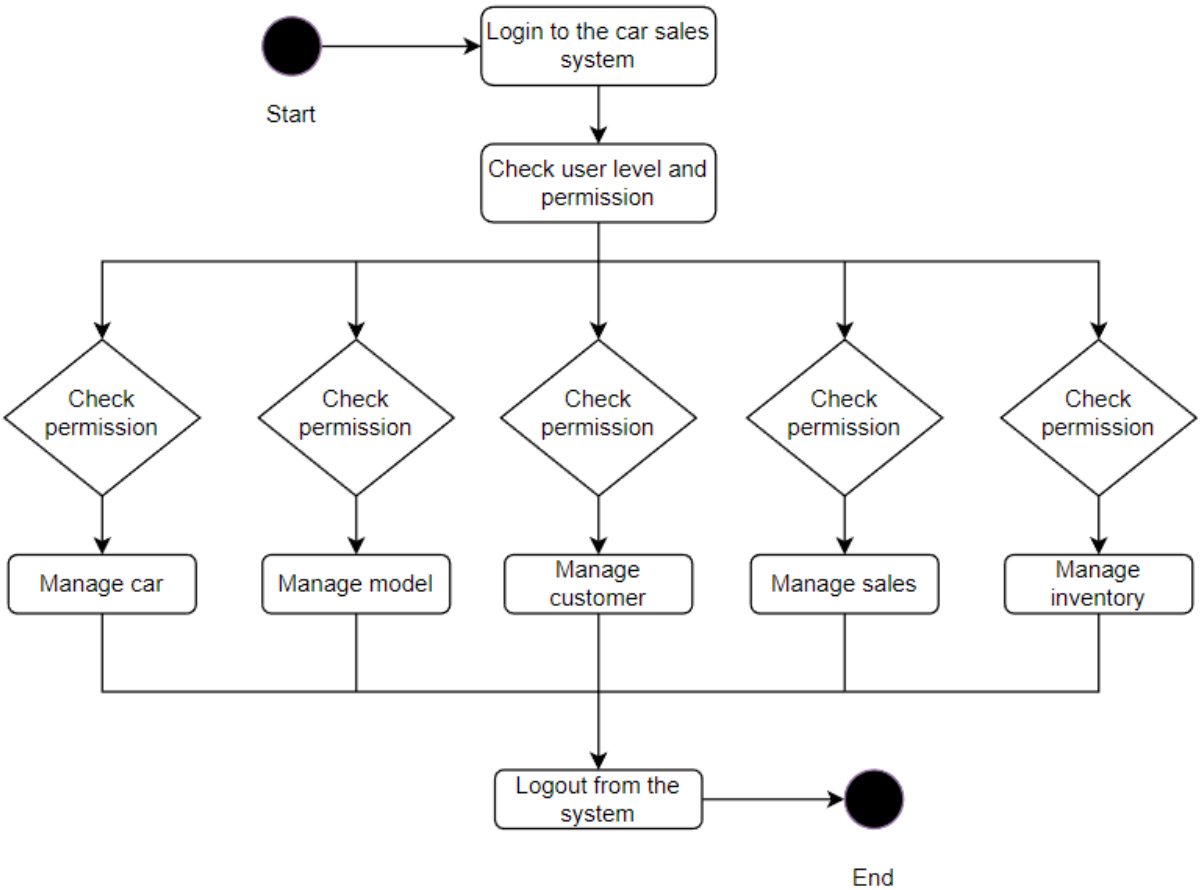


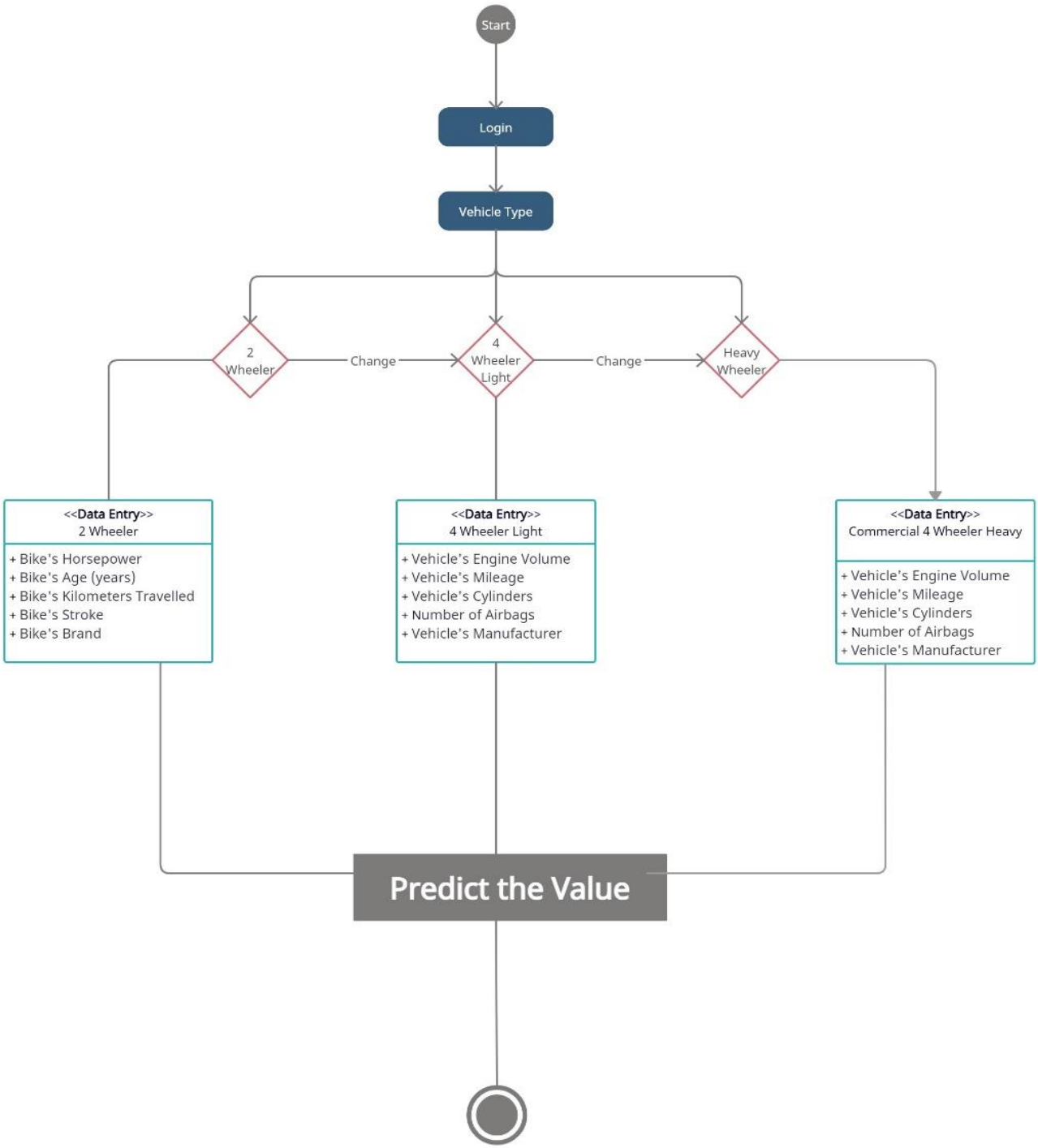


CHAPTER 4

UML Diagram






Start
Login
Vehicle Type
2 Wheeler
Change
4 Wheeler Light
Change
Heavy Wheeler
<<Data Entry>>
2 Wheeler
+ Bike's Horsepower
+ Bike's Age (years)
+ Bike's Kilometers Travelled
+ Bike's Stroke
+ Bike's Brand
<<Data Entry>>
4 Wheeler Light
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
<<Data Entry>>
Commercial 4 Wheeler Heavy
+ Vehicle's Engine Volume
+ Vehicle's Mileage
+ Vehicle's Cylinders
+ Number of Airbags
+ Vehicle's Manufacturer
Predict the Value

CHAPTER 5

Algorithm

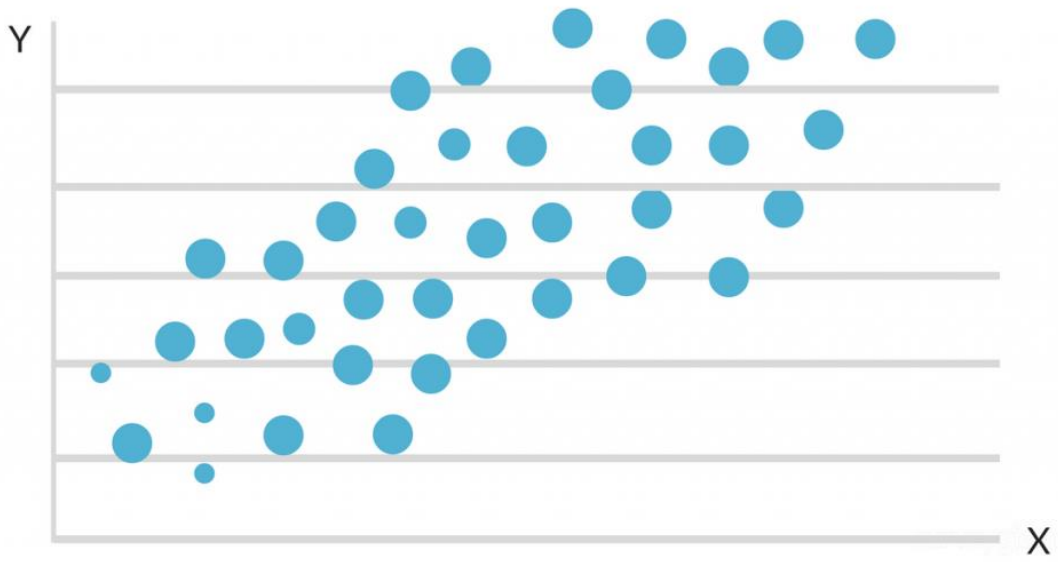
Regression analysis is a reliable method of identifying which variables have impact on a topic of interest. The process of performing a regression allows you to confidently determine which factors matter most, which factors can be ignored, and how these factors influence each other.

In order to understand regression analysis fully, it's essential to comprehend the following terms:

Dependent Variable: This is the main factor that you're trying to understand or predict.

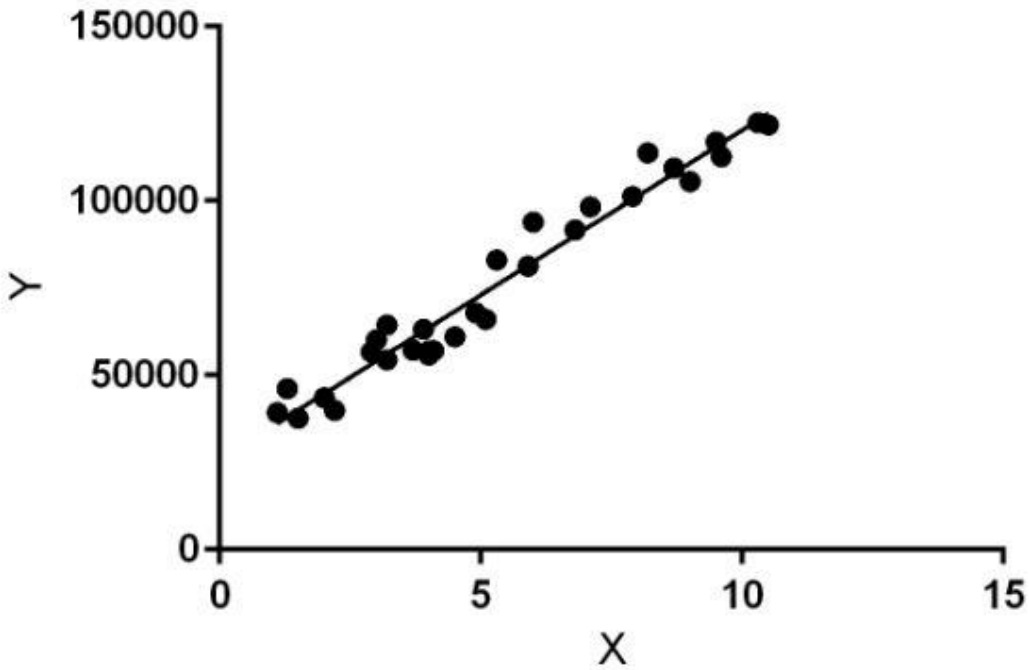
Independent Variables: These are the factors that you hypothesize have an impact on your dependent variable.

In our application training example above, attendees' satisfaction with the event is our dependent variable. The topics covered, length of sessions, food provided, and the cost of a ticket are our independent variables.



i. Liner Regression

A machine learning algorithm based on supervised learning is linear regression. It executes a regression operation. Regression uses independent variables to model a goal prediction value. It is mostly used to determine how variables and forecasting relate to one another. The type of link between dependent and independent variables that different regression models take into account, as well as the quantity of independent variables utilised, are what make them different..



The job of predicting a dependent variable's value (y) based on an independent variable is carried out using linear regression (x). Therefore, x (the input) and y (the output) are found to be linearly related using this regression approach (output). Thus, the term "linear regression" was coined.

In the diagram above, X represents a person's job history and Y represents their wage. The regression line fits our model the best.

Hypothesis function for Linear Regression:

$$y = \theta_1 + \theta_2.x$$

As we train the provided model:

input training data (one input variable, or parameter), x

Labels to data, y (supervised learning)

The model fits the best line to predict the value of y for a given value of x during training. By locating the best 1 and 2 values, the model produces the best regression fit line.

One: interception

2: x's coefficient

The best fit line is obtained after determining the best 1 and 2 values. As a result, when we use our model to make predictions, it will do so by predicting the value of y given the value of x as an input.

Cost Function (J):

By obtaining the best-fit regression line, the model seeks to forecast y values with the smallest possible error between the predictions and the actual values. To find the optimal value that minimises the error between the predicted y value (pred) and real y value, it is crucial to change the 1 and 2 values (y)

$$minimize \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

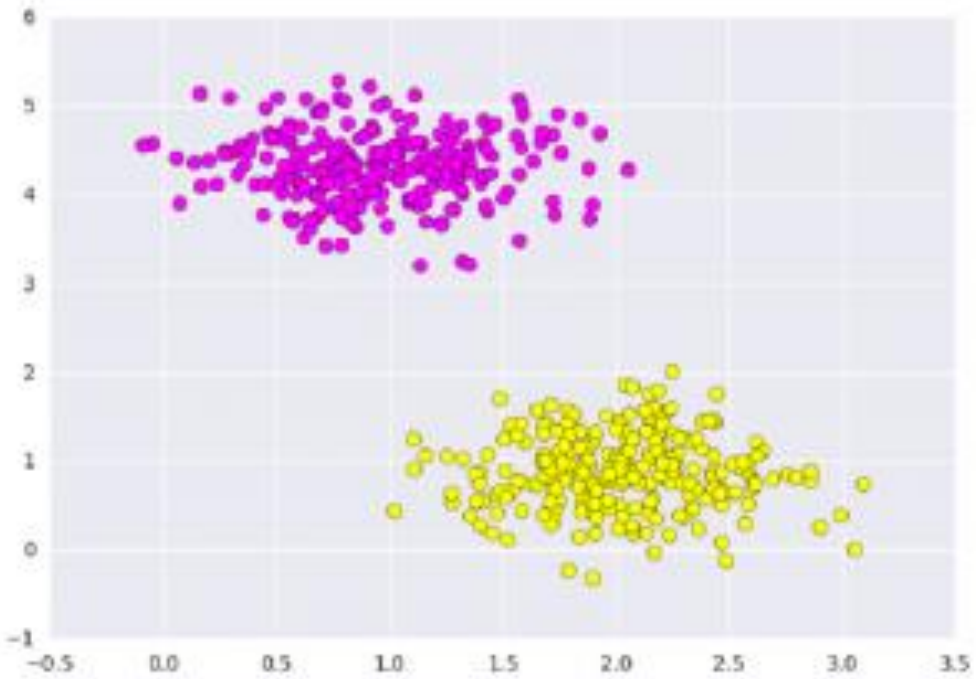
$$J = \frac{1}{n} \sum_{i=1}^{n} (pred_i - y_i)^2$$

Cost function(J) of Linear Regression is the Root Mean Squared Error (RMSE) between predicted y value (pred) and true y value (y).

ii. Support Vector Machine

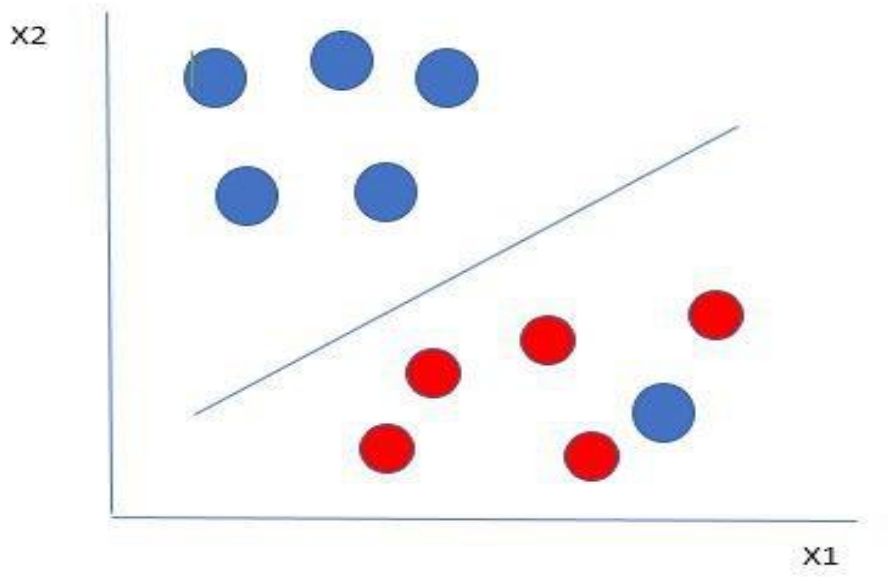
In machine learning, support vector machines (SVMs, also support vector networks) are supervised learning models with associated learning algorithms that analyze data used for classification and regression analysis. A Support Vector Machine (SVM) is a discriminative classifier formally defined by a separating hyperplane. In other words, given labeled training data (supervised learning), the algorithm outputs an optimal hyperplane which categorizes new examples.

An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. In addition to performing linear classification, SVMs can efficiently perform a non-linear classification, implicitly mapping their inputs into high-dimensional feature spaces.



A supervised machine learning approach called Support Vector Machine (SVM) is utilised for both classification and regression. Although we often refer to regression concerns, categorization is the most appropriate term. Finding a hyperplane in an N-dimensional space that clearly classifies the data points is the goal of the SVM method. The number of features determines the hyperplane's size. The hyperplane is essentially a line if there are just two input characteristics. The hyperplane turns into a 2-D plane if there are three input characteristics. Imagining something with

more than three characteristics gets challenging.



Modules Used:

Module 1: Data Integration

Data from a excel is integrated to the machine for data analysis. This data is then pre-processed using LabelEncoder(). Data of vehicles with attributes like engine volume, Mileage, Distance travelled, Horse Power, Stokes, Airbags, Bike company are integrated in to representing as a graph.

Module 2: ML Algorithm

Using python libraries, we integrate ML algorithms like SVM, Liner regression and many others to analysis the data. In this project we used only liner but other algorithms are also possible in which we can predict the values but making a slight change in the source code.

Module 3: **Value prediction and visualization**

In this module we have data visualization when in graphs are being introduced to show relation between the attributes but plotting the graphs using matplotlib library. And using ML we predict the value to be paid as well. This prediction is selected by analysis of the data in the CSV files we integrated.

Python Libraries Used:

- **Streamlit** is an open-source app framework in Python language. It helps us create web apps for data science and machine learning in a short time.
- **Pandas** is an open-source Python package that is most widely used for data science/data analysis and machine learning tasks.
- **NumPy** is a Python library used for working with arrays. It also has functions for working in domain of linear algebra, Fourier transform, and matrices.
- **Matplotlib** is a comprehensive library for creating static, animated, and interactive visualizations in Python.
- **Pyplot** is an API (Application Programming Interface) for Python's matplotlib that effectively makes matplotlib a viable open-source alternative to MATLAB
- **Sklearn** includes many effective tools for statistical modelling and machine learning, including dimensionality reduction, classification, regression, and clustering.

CHAPTER 6

Planning & Analysis

One of the most common types of non-life insurance is automobile insurance. This is due to the fact that it is legally required throughout the majority of the world. Every motor vehicle must be registered with the road transportation authority and have third-party liability insurance. The fundamental tenet is that a motor vehicle could result in property damage or bodily harm, necessitating insurance.

A legislative requirement under the Motor Vehicles Act is third-party liability coverage, generally known as motor third-party insurance or "act only" coverage. Since someone other than the two parties to the contract is the policy's beneficiary, it is known as a "third-party" cover (the car owner and the insurance company). The insured receives no benefits from the policy. It does, however, provide coverage for the insured's legal responsibility for loss or damage to third party property or death or disabling of a third party.

CHAPTER 7

Designing and Product Architecture

Analyze the information such as distance travelled since the time it is active, years used, model of car, and other factors to estimate the insurance amount to be paid and give maintenance alerts if required.

Give an option to the customer whether they like to order a technician to do the work or the customer themselves do it if they have the knowledge, however choosing the later will not give any warranty or free replacement if anything goes wrong.

Estimate the insurance claim that would be received by the customer due to faults caused by some unforeseen circumstances.

The system will also provide e-commerce support where in, if the issue is with a particular part, then that particular part will be searched online and displayed to the customer by using the NLP model.

CHAPTER 8

Implementation

Historically, technicians were called in order to assess the problems which the car has and they would suggest the estimated cost for fixing the issue as well.

In that system the chances of scamming the customers, not properly fixing the fault, not paying the insurance money to the customer due to lack of evidence and asking money way too much than the actual amount was common.

E-vehicle health monitoring and commerce systems were thus introduced to combat this problem from the market and provide a fair assessment for the problem at the best cost.

The current vehicle health monitoring systems uses ML algorithms to determine the condition of the vehicle and provide diagnosis for the same. The replacement for the faulty parts is provided by the system at a minimum cost which is then installed by the technician.

If however the fault is there due to some unforeseen circumstances, diagnosis is done and the insurance money is paid to the customer.

CHAPTER 9

Developing

Following along with the developmental model fitting for this project, webpage representation was deemed a good fit for prototyping and representing the key functionalities of the product and its corresponding modules. HTML, CSS, Python and SQL languages have been used for making the frontend and the backend.

In addition to this, machine learning algorithms like Support Vector Machines and Linear Regression have been used in order to perform insurance prediction for the vehicle. In order to make the web application stream lit was used.

With the use of module and a third party deployer, it was made possible for us to deploy a basic functioning website to analysis inputs and provide output for given data

CHAPTER 10

Testing

The primary function of testing is to detect bugs. The scope of testing includes execution of that code in various environments and also to examine the aspects of the code. The basic purpose of testing is verification and validation in order to find various errors and problems and aim of fixing them. It checks and tests of items, which includes software, performance and consistency of software by evaluating the results against pre-defined requirements. Verification is done order to check whether the program is working as intended or not and validation is performed to see whether all the systems is working fine or not. Testing activities also provides the chance to review requirements for important quality attributes. There are many activities that are performed during the testing.

These are:

- Test Analysis
- Test Design
- Test Execution 32

These activities are required to reduce the rework which results in reducing the cost and time. Software testing is an ongoing process which we can't stop in between.

Testing is required in SDLC due to the following reasons:

- To identify the errors
- To remove ambiguity
- To improve the reputation of the company
- To improve quality of the product
- To remove Hazards
- For verification and validation
- To improve reliability
- To improve cost
- To increase the usability

CHAPTER 11

Maintenance

It entails setting up the website on the computer system and starting the main operations. In this stage, users are introduced to the website for the first time and given the opportunity to interact with it. It entails keeping the website constantly updated to make sure it complies with the most recent updates to the same and current information needs. The developer evaluates the modifications and incorporates them as well. The following tasks are included in the implementation phase. the creation of websites, limit which people may make changes to the website, and tracking logs to maintain tabs on data and website activity.

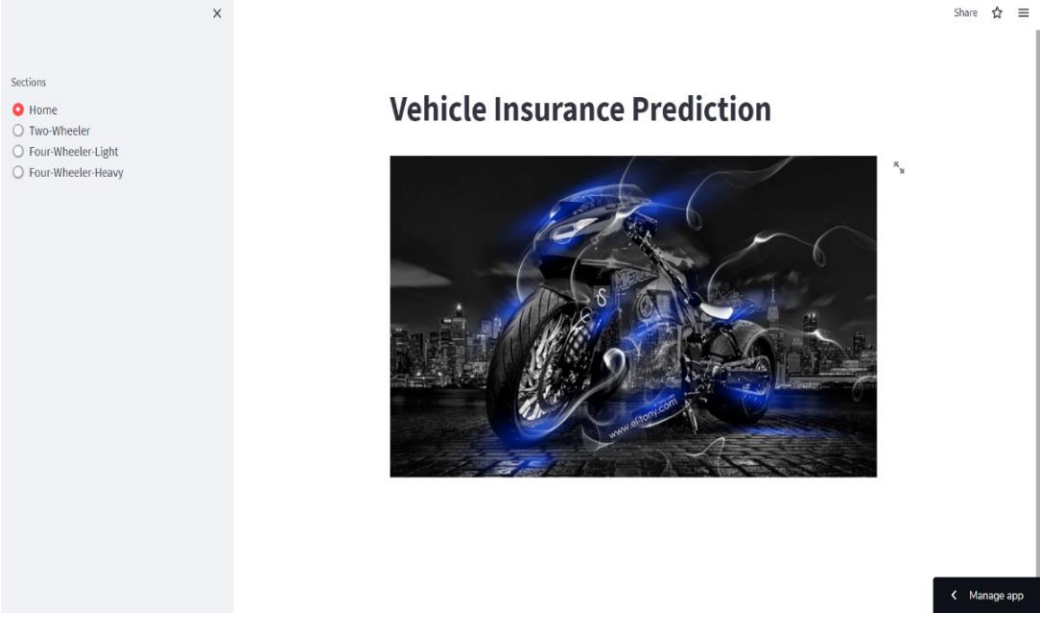
The WEB DEVELOPMENT LIFE CYCLE includes these phases, which take into account the following 10 guidelines:

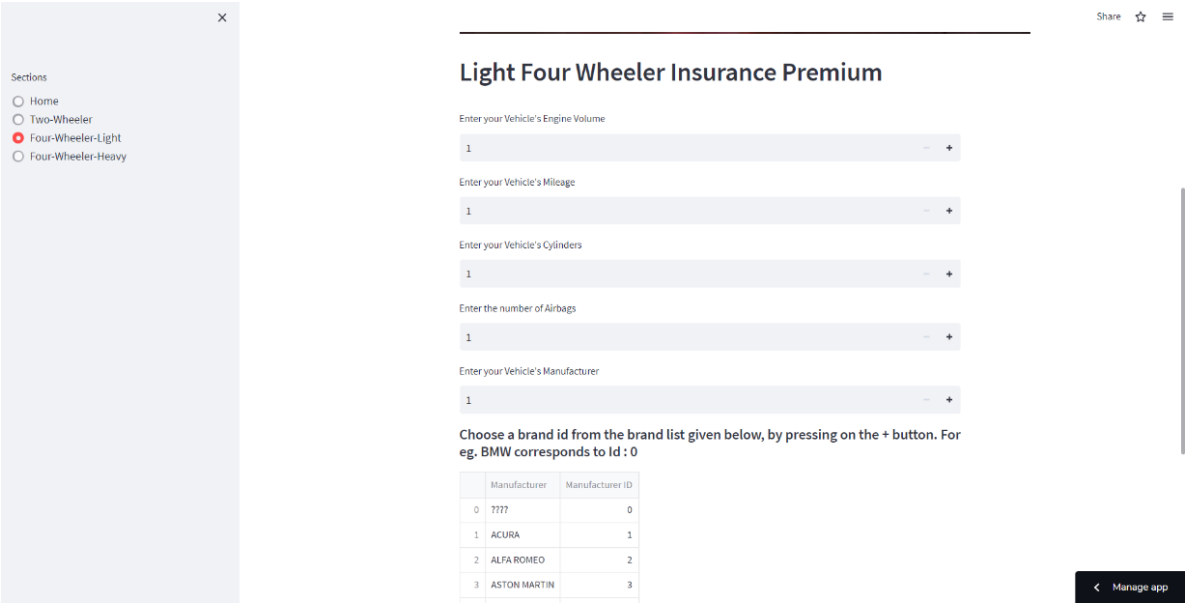
- Rule No. 1: Website Goals To get the intended results, the primary goal for the website's construction must be determined and evaluated.
- Rule 2: Website Map Prepare the overall website's physical look as well as its modularized structure. This might serve as a blueprint for future work.
- Rule 3: Gathering Requirements Include all client requirements and requirements criteria. The development process would be more succinct the more explicit the requirements are.

- Rule 4: Verification Before launching the website, the needs that were gathered and the design criteria needed to be checked against intellectual property laws to prevent privacy concerns.
- Rule 5: Website designs should be straightforward, clear of ambiguity, and uncomplicated so that the general public may readily understand the material.
- Rule 6: Avoid shortcuts Offer readers navigational tools to make it easier for them to browse the website quickly and get the information they need.
- Rule 7: Unification After the structure is created, the website may be evaluated against other similarly functional websites to provide the user a special application experience.
- Rule 8: Multilingualism The extra features can be added to give multilingual, handicapped consumers with technical help.
- Rule 9: Testing After doing multiple rounds of testing, the developed web application should be published. The testing verifies that the completed system meets the initial objectives set out.
- Rule 10: Upkeep The website must be updated often to keep it accurate and of high caliber. Additionally, the consumer may receive current information from this.

CHAPTER 12

Screenshots





Two Wheeler Insurance Premium

Enter your Bike's Horsepower

200

Enter your Bike's Age (years)

2

Enter your Bike's Kilometers Travelled (in 1000 km)

2

Enter your Bike's Stroke

2

Enter your Bike's Brand

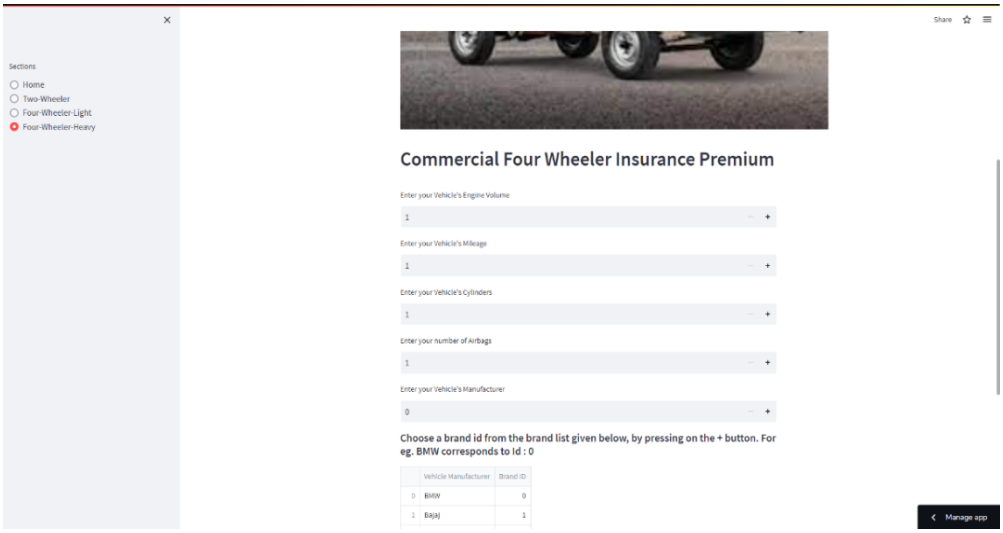
12

Choose a brand id from the brand list given below, by pressing on the + button. For eg. BMW corresponds to Id : 0

	Bike Brand	Brand ID
0	BMW	0
1	Bajaj	1
2	Benelli	2
3	Ducati	3
4	Harley-Davidson	4
5	Hero	5
6	Honda	6
7	Hyosung	7
8	Ideal	8
9	Indian	9

Predict

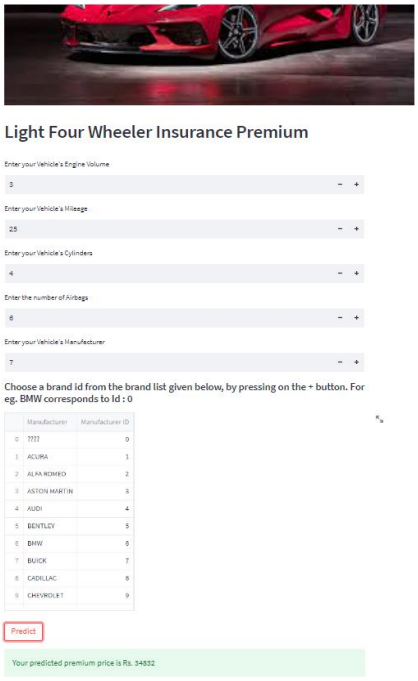
Your predicted premium price is Rs. 7833.1



130 lines (101 sloc) | 5.08 KB

```python
from numpy.core.frommumeric import size
import streamlit as st
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn import preprocessing

st.title("Vehicle Insurance Prediction")



nav = st.sidebar.radio("Sections",["Home","Two-Wheeler","Four-Wheeler-Light","Four-Wheeler-Heavy"])
if nav == "Home":
    st.image("bike.jpg",width= 700)




elif nav == "Two-Wheeler":

    data = pd.read_csv("bikes.csv")
    data1 = data.copy()
    label_encoder = preprocessing.LabelEncoder()
    data['brand']= label_encoder.fit_transform(data['brand'])
    data['brand'].unique()
    lr = LinearRegression()
    Z = data[['power', 'age', 'kms_driven','stroke','brand']]
    lr.fit(Z, data['price'])
    st.image("bike.jpg",width= 800)
    st.header("Two Wheeler Insurance Premium")
    val1 = st.number_input("Enter your Bike's Horsepower",100,2000,step = 100)
    val2 = st.number_input("Enter your Bike's Age (years)",1,40,step = 1)
    val3 = st.number_input("Enter your Bike's Kilometers Travelled (in 1000 km)",1,50,step = 1)
    val4 = st.number_input("Enter your Bike's Stroke",1,5,step = 1)
    val5 = st.number_input("Enter your Bike's Brand",0,22,step = 1)

    val = np.array([val1,val2,val3,val4,val5]).reshape(1,-1)

```

```python
    pred = lr.predict(val)[0]
    pred = round(pred)/(10)
    if st.button("Predict"):
        st.success(f"Your predicted premium price is Rs. {pred}")


elif nav == "Four-Wheeler-Light":

    data = pd.read_csv("car_price_prediction.csv")
    data1 = data.copy()
    label_encoder = preprocessing.LabelEncoder()
    data['Manufacturer']= label_encoder.fit_transform(data['Manufacturer'])
    data['Manufacturer'].unique()
    lr = LinearRegression()
    Z = data[['Engine volume','Milage','Cylinders','Airbags','Manufacturer']]
    lr.fit(Z, data['Price'])
    st.image("car.jpeg",width= 800)
    st.header("Light Four Wheeler Insurance Premium")
    val1 = st.number_input("Enter your Vehicle's Engine Volume",1,20,step = 1)
    val2 = st.number_input("Enter your Vehicle's Mileage",1,100,step = 1)
    val3 = st.number_input("Enter your Vehicle's Cylinders",1,8,step = 1)
    val4 = st.number_input("Enter the number of Airbags",1,20,step=1)
    val5 = st.number_input("Enter your Vehicle's Manufacturer",1,63,step = 1)

    val = np.array([val1,val2,val3,val4,val5]).reshape(1,-1)

    datax = [data1["Manufacturer"], data["Manufacturer"]]
    headers = ["data1", "data"]
    df3 = pd.concat(datax, axis=1, keys=headers)
    dfx = df3.sort_values('data')
    dfx = dfx.drop_duplicates()
    dfx.reset_index(drop=True, inplace=True)
    st.markdown("##### Choose a brand id from the brand list given below, by pressing on the + button. For eg. BMW corresponds to Id : 0")
    dfx.rename(columns={'data1': 'Manufacturer','data':'Manufacturer ID'}, inplace=True)
    st.dataframe(dfx)


    pred = lr.predict(val)[0]
    pred = round(pred)
    if st.button("Predict"):
        st.success(f"Your predicted premium price is Rs. {pred}")
```



Links-

- https://vehicleinsurancepredictor18vk.streamlit.app
- https://bikevaluepredictorvk18.streamlit.app

CHAPTER 13

Result & Discussion

Dataset for this project was a big problem as many sites did not have a proper arranged data sets for analysis. Some had name and value while some had distance travelled but no other attribute. In this process there was a point where insufficient data was available for the machine to work, wherein prediction was not as expected.

After a thorough search and time, we could find a dataset that could work with the analysis where it had horse power, milage, stokes and other import elements for a vehicle's engine

When it comes to results, with the use of liner regression it was always possible for negative values which implies they are not worth insurance or the vehicle health was bad.

These cases occurred frequently during the data analysis and prediction time which took a lot of time

Even though the value is predicted, with more data on vehicle and previous insurance issued for those along with extra attributes can really help the machine provide more accurate value

Also, use of other algorithms like SVM, logistic regression, Decision tree, KNN, K-mean

can provide more insights on this prediction of value.

Conclusion

The vehicle monitoring system offers accurate information on all fundamental factors influencing a vehicle's performance. Due to the early detection of damage made possible by this information, the performance of the vehicle will improve. As a result, it gives the owner accurate information so that during car servicing, the service centers cannot present incorrect information and demand additional payment. The vehicle insurance app aids the owner in obtaining the most insurance payout possible in the event that the vehicle develops flaws or malfunctions as a result of normal wear and tear. According to the company's insurance policy, the owner can obtain the greatest claim possible, and that too without deceit or fraud. This method of auto insurance aids in eradicating such offensive behaviors.

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Conference Publication

Publication process not yet started

Plagiarism Report

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5	Chapter 3: Block Diagram	0	0	1
6	Chapter 4: UML Diagram	0	0	1
7	Chapter 5: Algorithms	4	2	2
8	Chapter 6: Planning & Analysis	3	1	3
9	Chapter 7: Designing the Product Architecture	3	3	6
10	Chapter 8: Implementation	4	1	5
11	Chapter 9: Developing	1	4	4
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